

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Lecture 11: Evolution of the Reference Parameter Set

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Overview

Successive releases of the reference parameters are compared with a Mahalanobis distance and a fixed set of observables.

1 Physical assumptions

All coordinate conditions and boundary conditions are stated before a parameter is fitted.

$$(N_\theta, N_O)_{2026} = (47, 52)$$

2 Field equations

The first two displayed relations establish the notation used in the derivation.

$$d^2 = (\theta_{n+1} - \theta_n)^T C_n^{-1} (\theta_{n+1} - \theta_n)$$

3 Observable limit

Weak-field, static, and point-source limits are checked independently.

$$L = \sum_{a \in \text{Registry}} R_a^2$$

4 Parameter dependence

Parameter combinations are compared through derivatives of the reference observables.

5 Example

The calculation is repeated for one analytic case and one tabulated data point.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- SGT Historical Concordance
- Table Governance Standard 2026
- SGT9 issuance run log excerpt